

GAIRA V6 — Chemical Theme Reference

Thirteen chemical themes: definition, membership, evidence and failure modes

A chemical theme is a **grouping of MSS motifs**, not a grouping of components and not a biological state. Every theme below is a nameable biochemical class — that was enforced as a constraint, not hoped for.

chemical theme	member MSS motifs
Purine	oxopurine_carbonyl, purine_ring_breathing
Pyrimidine	pyrimidine_ring
Monosaccharide + Phosphate	glycan_co_network, nucleic_backbone_phosphate
Aromatic residue	aromatic_ring_residue
Protein backbone	protein_amide_backbone
Free amino acid	amino_acid_zwitterion
Acyl lipid	fatty_acyl_chain, triglyceride_ester
Sterol	sterol_ring_system
Polysaccharide	polysaccharide_glycosidic
Organic acid	carboxylate_organic_acid
Sulfur metabolite	sulfur_heterocycle_thione
Flavin + Porphyrin	flavin_redox_cofactor, porphyrin_macrocycle
Carotenoid	carotenoid_polyene

Part 7 — full evaluation over every Raman grounding analyte

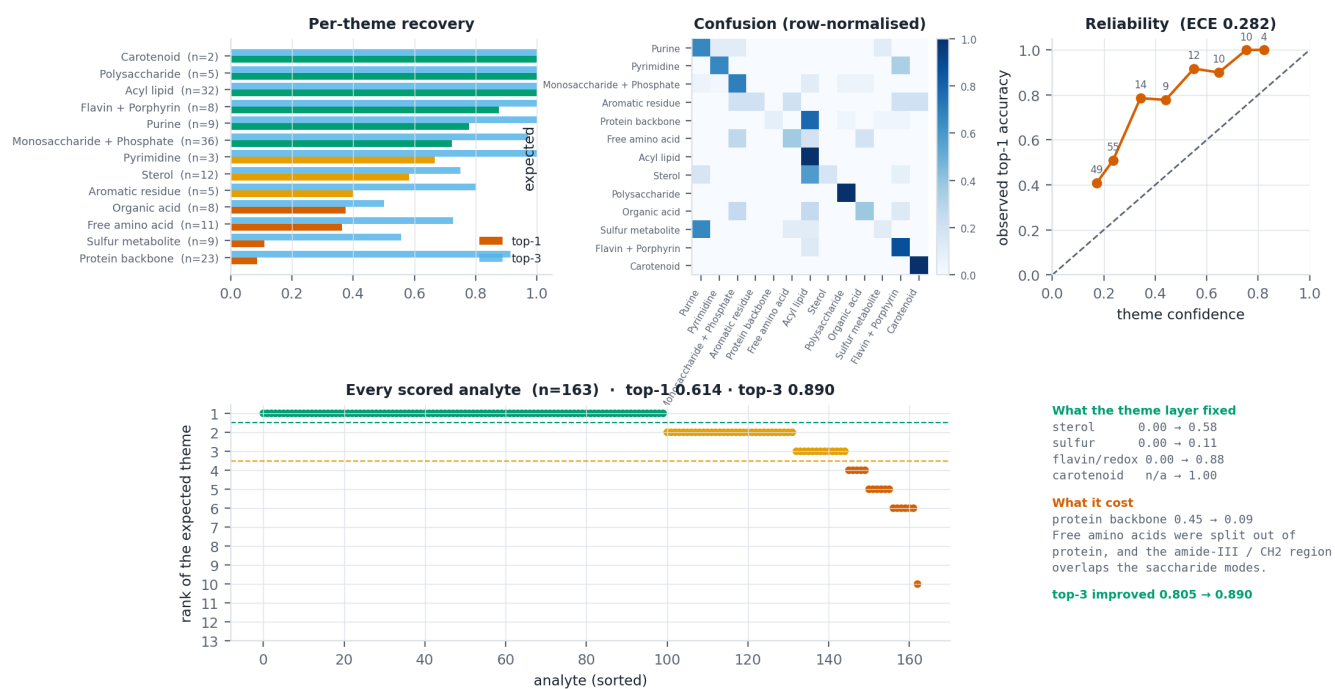


Figure 1 — Per-theme recovery, confusion and calibration over the full Raman corpus.

Purine

Definition

Spectral evidence for this theme is the summed activation of oxopurine_carbonyl, purine_ring_breathing. 2,6-dioxopurine ring + carbonyl modes shared by xanthine/urate/guanine. Ring-breathing and stretching modes of the purine scaffold (adenine-type).

Membership and evidence

included MSS motifs	oxopurine_carbonyl, purine_ring_breathing
key Raman bands (cm ⁻¹)	640, 720, 730, 938, 1230, 1250, 1330, 1334, 1480, 1580
representative components	c15, c3, c19, c23, c17, c20
representative analytes	a-dna, adenine, b-dna, creatinine, guanine, hypoxanthine
corpus analytes	9 (5.5% of the scored set)
motif confidence	0.691

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.778	1.000	1	0.273	Pyrimidine

Known failure cases

These analytes should express **Purine** but do not rank it first: *creatinine*, *urate*.

Pyrimidine

Definition

Spectral evidence for this theme is the summed activation of pyrimidine_ring. Ring modes of the pyrimidine scaffold (uracil / cytosine / thymine).

Membership and evidence

included MSS motifs	pyrimidine_ring
key Raman bands (cm ⁻¹)	560, 578, 790, 1236, 1396, 1660
representative components	c17, c23, c13, c7, c14, c21
representative analytes	cytosine, thymine, uracil
corpus analytes	3 (1.8% of the scored set)
motif confidence	0.487

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.667	1.000	1	0.317	Flavin + Porphyrin

Known failure cases

These analytes should express **Pyrimidine** but do not rank it first: *thymine*.

Monosaccharide + Phosphate

Definition

Spectral evidence for this theme is the summed activation of `glycan_co_network`, `nucleic_backbone_phosphate`. Carbohydrate C-O / C-C / C-O-C ring and anomeric modes.

Membership and evidence

included MSS motifs	glycan_co_network, nucleic_backbone_phosphate
key Raman bands (cm ⁻¹)	788, 816, 835, 872, 916, 920, 1052, 1054, 1094, 1108
representative components	c17, c10, c4, c23, c18, c20
representative analytes	(+)-arabinose, (+)-dextrose, (+)-fucose, (+)-galactosamine, (+)-galactose, (+)-glucose
corpus analytes	36 (22.1% of the scored set)
motif confidence	0.607

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.722	0.972	1	0.253	Acyl lipid

Known failure cases

These analytes should express **Monosaccharide + Phosphate** but do not rank it first: (+)-galactose, (+)-xylose, glucosamine, glucose oxidase, phosphate, phosphatidylinositol.

Aromatic residue

Definition

Spectral evidence for this theme is the summed activation of `aromatic_ring_residue`. Aromatic side-chain ring modes — phenylalanine 1004 cm⁻¹ ring breathing, tyrosine 828/850 Fermi doublet, tryptophan ring modes.

Membership and evidence

included MSS motifs	aromatic_ring_residue
key Raman bands (cm ⁻¹)	640, 828, 852, 1006, 1178, 1614
representative components	c5, c6, c11, c15, c0, c22
representative analytes	alanine, histidine, phenylalanine, tryptophan, tyrosine
corpus analytes	5 (3.1% of the scored set)
motif confidence	0.601

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.400	0.800	2	0.255	Monosaccharide + Phosphate

Known failure cases

These analytes should express **Aromatic residue** but do not rank it first: *histidine, phenylalanine, tryptophan*.

Protein backbone

Definition

Spectral evidence for this theme is the summed activation of `protein_amide_backbone`. Peptide backbone signature — amide I (~1650), amide III (~1240), C α -C and backbone skeletal modes.

Membership and evidence

included MSS motifs	protein_amide_backbone
key Raman bands (cm ⁻¹)	856, 1000, 1130, 1240, 1336, 1654
representative components	c2, c10, c12, c18, c6, c14
representative analytes	alb, albumin, carbonic anhydrase, collagen, elastase, elastin
corpus analytes	23 (14.1% of the scored set)
motif confidence	0.731

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.087	0.913	2	0.203	Acyl lipid

Known failure cases

These analytes should express **Protein backbone** but do not rank it first: *alb, albumin, carbonic anhydrase, collagen, elastase, elastin*.

LOW RELIABILITY. Protein backbone is ranked first for only 9% of the analytes that should express it, in-domain. Do not report it as a finding.

Free amino acid

Definition

Spectral evidence for this theme is the summed activation of `amino_acid_zwitterion`.

Membership and evidence

included MSS motifs	amino_acid_zwitterion
key Raman bands (cm ⁻¹)	870, 930, 1030, 1345, 1410, 1520
representative components	c22, c14, c9, c11, c21, c10
representative analytes	arginine, asparagine, aspartic acid, glutamate, glycine, hydroxyproline
corpus analytes	11 (6.7% of the scored set)
motif confidence	0.599

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.364	0.727	2	0.209	Monosaccharide + Phosphate

Known failure cases

These analytes should express **Free amino acid** but do not rank it first: *arginine, asparagine, aspartic acid, isoleucine, leucine, proline*.

LOW RELIABILITY. Free amino acid is ranked first for only 36% of the analytes that should express it, in-domain. Do not report it as a finding.

Acyl lipid

Definition

Spectral evidence for this theme is the summed activation of fatty_acyl_chain, triglyceride_ester.

Membership and evidence

included MSS motifs	fatty_acyl_chain, triglyceride_ester
key Raman bands (cm ⁻¹)	1062, 1065, 1128, 1160, 1265, 1301, 1440, 1655, 1745
representative components	c16, c7, c1, c12, c4, c2
representative analytes	12-methyltetradecanoic acid, 13-methylmyristic acid, 14-methylhexadecanoic acid, 14-methylpentadecanoic acid, 15-methylpalmitic acid, arachidic acid
corpus analytes	32 (19.6% of the scored set)
motif confidence	0.630

Performance

top-1	top-3	median rank	mean confidence	most confused with
1.000	1.000	1	0.669	-

Sterol

Definition

Spectral evidence for this theme is the summed activation of sterol_ring_system. Fused steroid ring skeletal modes (cholesterol / estrone family).

Membership and evidence

included MSS motifs	sterol_ring_system
key Raman bands (cm ⁻¹)	548, 608, 702, 958, 1128, 1440, 1670
representative components	c3, c8, c13, c12, c2, c20
representative analytes	cholesterol, cholesteryl linoleate, cholesteryl oleate, cholesteryl palmitate, cholesteryl stearate, diethylstilbestrol
corpus analytes	12 (7.4% of the scored set)

motif confidence	0.618
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Performance

top-1	top-3	median rank	mean confidence	most confused with
0.583	0.750	1	0.329	Acyl lipid

Known failure cases

These analytes should express **Sterol** but do not rank it first: *cholesterol*, *cholesteryl palmitate*, *diethylstilbestrol*, *estrone*, *ethinylestradiol*.

Polysaccharide

Definition

Spectral evidence for this theme is the summed activation of polysaccharide_glycosidic.

Membership and evidence

included MSS motifs	polysaccharide_glycosidic
key Raman bands (cm ⁻¹)	890, 940, 1080, 1122, 1340, 1460
representative components	c18, c9, c14, c2, c16, c10
representative analytes	amylopectin, amylose, cellulose, chitin, glycogen
corpus analytes	5 (3.1% of the scored set)
motif confidence	0.507

Performance

top-1	top-3	median rank	mean confidence	most confused with
1.000	1.000	1	0.238	-

Organic acid

Definition

Spectral evidence for this theme is the summed activation of carboxylate_organic_acid. Carboxylate symmetric stretch (~1400) and C-COO / skeletal modes of small TCA-cycle and metabolic acids.

Membership and evidence

included MSS motifs	carboxylate_organic_acid
key Raman bands (cm ⁻¹)	760, 940, 1382, 1398, 1628
representative components	c23, c14, c22, c5, c18, c17
representative analytes	acetoacetate, ascorbate, aspartate, citrate, fumarate, lactate
corpus analytes	8 (4.9% of the scored set)
motif confidence	0.548

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.375	0.500	3	0.228	Monosaccharide + Phosphate

Known failure cases

These analytes should express **Organic acid** but do not rank it first: *ascorbate*, *aspartate*, *fumarate*, *lactate*, *malic acid*.

LOW RELIABILITY. Organic acid is ranked first for only 38% of the analytes that should express it, in-domain. Do not report it as a finding.

Sulfur metabolite

Definition

Spectral evidence for this theme is the summed activation of sulfur_heterocycle_thione. Thione / thiol and C-S modes — the ergothioneine thione ring and cysteine/ glutathione thiols.

Membership and evidence

included MSS motifs	sulfur_heterocycle_thione
key Raman bands (cm ⁻¹)	536, 640, 674, 838, 970, 1200
representative components	c19, c3, c15, c21, c20, c5
representative analytes	acetyl coenzyme a, acetyl-coa, coenzyme a, cysteine, ergothioneine, glutathione
corpus analytes	9 (5.5% of the scored set)
motif confidence	0.726

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.111	0.556	3	0.191	Purine

Known failure cases

These analytes should express **Sulfur metabolite** but do not rank it first: *acetyl coenzyme a*, *acetyl-coa*, *coenzyme a*, *cysteine*, *glutathione*, *glutathione transferase*.

LOW RELIABILITY. Sulfur metabolite is ranked first for only 11% of the analytes that should express it, in-domain. Do not report it as a finding.

Flavin + Porphyrin

Definition

Spectral evidence for this theme is the summed activation of flavin_redox_cofactor, porphyrin_macrocycle. Isoalloxazine (flavin) and related redox-cofactor ring modes — riboflavin/ FAD, ubiquinone. Resonance-enhanced tetrapyrrole macrocycle modes of heme proteins (hemoglobin, myoglobin, cytochrome c) — pyrrole breathing and C α -C μ stretches.

Membership and evidence

included MSS motifs	flavin_redox_cofactor, porphyrin_macrocycle
key Raman bands (cm ⁻¹)	750, 754, 1130, 1178, 1348, 1362, 1408, 1578, 1590, 1620
representative components	c0, c8, c13, c22, c17, c2
representative analytes	cytochrome c, ferritin, hemoglobin, horseradish peroxidase, melanin, myoglobin
corpus analytes	8 (4.9% of the scored set)
motif confidence	0.495

Performance

top-1	top-3	median rank	mean confidence	most confused with
0.875	1.000	1	0.352	Acyl lipid

Known failure cases

These analytes should express **Flavin + Porphyrin** but do not rank it first: *ferritin*.

Carotenoid

Definition

Spectral evidence for this theme is the summed activation of carotenoid_polyene.

Membership and evidence

included MSS motifs	carotenoid_polyene
key Raman bands (cm ⁻¹)	1005, 1157, 1190, 1270, 1520
representative components	c6, c12, c5, c4
representative analytes	carotene, β -carotene
corpus analytes	2 (1.2% of the scored set)
motif confidence	0.506

Performance

top-1	top-3	median rank	mean confidence	most confused with
1.000	1.000	1	0.184	-

Cross-theme notes

- **Acyl lipid, Polysaccharide and Carotenoid are perfect (1.00)** in-domain, but carotenoid rests on two analytes and should not be read as strong evidence.
- **Protein backbone (0.09) and Sulfur metabolite (0.11) are the weakest.** Both lose to Acyl lipid and Monosaccharide + Phosphate, whose motifs carry more component mass in the shared 1240–1460 cm⁻¹ region.
- **Sterol (0.58) is recovered for the first time** — V1's sterol theme never ranked first for any analyte — but the frozen atlas still has no component that isolates the steroid ring system.

- **Flavin + Porphyrin (0.88)** looks strong but rests on 8 analytes and two motifs that both borrow protein components; treat it as provisional.

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